

Proposal

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```
library(readr)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v purrr      1.1.0
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(dplyr)
data <- read_csv("data/powerlifting.csv")
```

Using Open Powerlifting data from past IPF World University Powerlifting Cups, we will look into the following research questions:

1. How does a collegiate athlete's body weight relate to the total weight that they lifted in their competition? We will look into whether heavier collegiate athletes achieve higher total lifts and how strong the correlation is between body weight and total weight lifted.
2. How does country, location of origin, relate to the total weight they lift at a competition? We will look into if collegiate athletes from certain locations exhibit significant differences in total lifted weight while keeping body weight the same.
3. How correlated are individual lifts (squat, bench, deadlift) to each other? Specifically, we will see how a collegiate athlete's performance in one lift predicts their performance in another. We will then finalize which lifts share the strongest relationship; therefore using one individual lift weight, we can predict another.

Why?

The research is valuable to collegiate powerlifters and coaches. When they are able to understand how bodyweight and country of origin correlate to total weight lifted and how the performance of individual lifts correlate to others, they can help these athletes set specific goals.

These are all important topics to dive into because of the popularity powerlifting is gaining in the world. For example, should one gain weight to reach a heavier lift weight? Also, are there differences in weightlifting habits based on cultural aspects? If so, more research can be done on food, training habits, or lifestyles of that country's culture. The acknowledgement of these topics could help adjust powerlifters' training, helping maximize their performance.

Our Data

Our dataset is from International Powerlifting Federation's World University Powerlifting Cup. We want to look at data from the most recent 5 years, from 2021-2025. Each row represents an athlete, with information about them and their performance at this respective meet. Each column represents different categorical and numerical variables: sex, age, bodyweight, individual weight lifted by lift, total weight lifted, DOT score, and country of origin are some.

Using this data, we can filter and run linear regression on our selected variables such as total lifted weight, body weight, country, and more. We are also able to conduct hypothesis testing and gauge model effectiveness using our data.

```
#|label: view-data
```

```
glimpse(data)
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```
Rows: 1,154
```

```
Columns: 43
```

```
$ Name      <chr> "Iurii Aydın", "Ahmet Yiğit Yılmaz", "Arif Mohammad S~
$ Sex       <chr> "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M"~
$ Event     <chr> "SBD", "SBD", "SBD", "SBD", "SBD", "SBD", "SBD", "SBD", "SBD"~
$ Equipment <chr> "Raw", "Raw", "Raw", "Raw", "Raw", "Raw", "Raw", "Raw", "Raw"~
$ Age       <dbl> 16.5, 18.0, 18.5, 19.0, 19.0, 20.0, 20.5, 21.0, 21.0, ~
$ AgeClass  <chr> "16-17", "18-19", "18-19", "18-19", "18-19", "18-19", "20-23", ~
$ BirthYearClass <chr> "14-18", "14-18", "19-23", "19-23", "19-23", "19-23", "19-23", ~
$ Division  <chr> "Open", "Open", "Open", "Open", "Open", "Open", "Open", "Open"~
$ BodyweightKg <dbl> 88.25, 65.55, 65.30, 72.95, 128.65, 81.60, 81.40, 92.~
$ WeightClassKg <chr> "93", "66", "66", "74", "120+", "83", "83", "93", "10~
$ Squat1Kg  <dbl> 190.0, 135.0, 127.5, 222.5, 195.0, 225.0, 182.5, 230.~
```

\$ Squat2Kg	<dbl> 202.5, 150.0, 137.5, 235.0, 210.0, 237.5, 195.0, -242~
\$ Squat3Kg	<dbl> 210.0, -167.5, -145.0, 240.0, 217.5, -242.5, -200.0, ~
\$ Squat4Kg	<lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
\$ Best3SquatKg	<dbl> 210.0, 150.0, 137.5, 240.0, 217.5, 237.5, 195.0, 230.~
\$ Bench1Kg	<dbl> 112.5, 140.0, 105.0, 132.5, 140.0, 127.5, 125.0, 127.~
\$ Bench2Kg	<dbl> 120.0, 145.0, 107.5, 142.5, -147.5, 135.0, 135.0, 135~
\$ Bench3Kg	<dbl> -125.0, 150.0, -115.0, 147.5, -147.5, 140.0, 140.0, 1~
\$ Bench4Kg	<lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
\$ Best3BenchKg	<dbl> 120.0, 150.0, 107.5, 147.5, 140.0, 140.0, 140.0, 140.~
\$ Deadlift1Kg	<dbl> 250.0, 180.0, 160.0, -285.0, 195.0, 240.0, 215.0, 260~
\$ Deadlift2Kg	<dbl> 267.5, 187.5, 175.0, -297.5, 210.0, 250.0, 232.5, 277~
\$ Deadlift3Kg	<dbl> -277.5, 192.5, 185.0, 307.5, 220.0, -262.5, -237.5, --
\$ Deadlift4Kg	<lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
\$ Best3DeadliftKg	<dbl> 267.5, 192.5, 185.0, 307.5, 220.0, 250.0, 232.5, 277.~
\$ TotalKg	<dbl> 597.5, 492.5, 430.0, 695.0, 577.5, 627.5, 567.5, 647.~
\$ Place	<chr> "18", "11", "12", "3", "4", "17", "21", "15", "12", "~
\$ Dots	<dbl> 390.24, 387.86, 339.60, 507.67, 324.39, 427.73, 387.3~
\$ Wilks	<dbl> 385.35, 388.90, 340.62, 505.07, 327.25, 423.16, 383.2~
\$ Glossbrenner	<dbl> 369.64, 378.02, 331.16, 488.60, 312.80, 407.37, 369.0~
\$ Goodlift	<dbl> 80.21, 77.05, 67.41, 102.79, 65.16, 87.61, 79.34, 85.~
\$ Tested	<chr> "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes~
\$ Country	<chr> "Turkey", "Turkey", "Qatar", "Turkey", "Turkey", "Ire~
\$ State	<lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
\$ Federation	<chr> "IPF", "IPF", "IPF", "IPF", "IPF", "IPF", "IPF", "IPF~
\$ ParentFederation	<chr> "IPF", "IPF", "IPF", "IPF", "IPF", "IPF", "IPF", "IPF~
\$ Date	<chr> "2025-07-16", "2025-07-16", "2025-07-16", "2025-07-16~
\$ MeetCountry	<chr> "Turkey", "Turkey", "Turkey", "Turkey", "Turkey", "Tu~
\$ MeetState	<lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
\$ MeetTown	<chr> "Istanbul", "Istanbul", "Istanbul", "Istanbul", "Ista~
\$ MeetName	<chr> "World University Powerlifting Cup", "World Universit~
\$ Sanctioned	<chr> "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes~
\$ Year	<int> 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025, ~

How?

1. How does a collegiate athlete's body weight relate to the total weight that they lifted in their competition?:

Our outcome variable is the total weight lifted, and the predictor is the athlete's body weight. We can run a single linear regression for this continuous outcome.

2. How does country, location of origin, relate to the total weight they lift at a competition?:

Our outcome variable is the total weight lifted, and the predictor is the athlete's body weight and country. We can run multiple linear regressions for this continuous outcome with the new predictor variable country to gauge the best combination of predictors.

3. How correlated are individual lifts (squat, bench, deadlift) to each other?:

Our outcome variable is the weight lifted for a certain lift, and the predictor is the weight lifted for another lift. For example, we can predict a bench weight using a squat weight. We can run a single linear regression for this continuous outcome.